

CSC 303

Database Management System

Examination: Semester Examination	Semester: Second Semester
Year: 2024	Duration: 2 hours
Total Questions: 5	

Instructions: Answer ALL questions. All questions carry equal marks unless otherwise stated.

Question 1

- (a) Define a Database Management System (DBMS) and explain its advantages over file-based systems. (8 marks)
- (b) Explain the three-schema architecture of DBMS with a diagram. (7 marks)

Question 2

- (a) What is normalization? Explain 1NF, 2NF, and 3NF with examples. (12 marks)
- (b) Why is normalization important in database design? (3 marks)

Question 3

Consider the following relational schema:

Student(matricNo, name, deptCode, level)

Department(deptCode, deptName, faculty)

Course(courseCode, courseTitle, creditUnit)

Write SQL queries to:

- (a) List all students in the Computer Science department. (5 marks)
- (b) Find the total number of students in each department. (5 marks)
- (c) Update the level of all 300-level students to 400. (5 marks)

— End of Examination —